

INTERNSHIP PRESENTATION

BEAST OS

A Cross-Platform EdTech Operating System for ICSE Board Preparation

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Internship & Project Overview

From internship responsibilities to production-facing front-end modules

Organisation

Beast Learners — an EdTech organisation building technology-driven solutions for school-level board examination preparation.

Role

Web Developer Intern with primary responsibility for the student-facing web interface and internal analytics dashboard.

Core Project

Beast OS — a unified platform combining syllabus tracking, AI tutoring, assessments and progress analytics.

- Internship focus: front-end/UI engineering
- Five major production-facing modules were engineered
- Work covered responsive layouts, reusable components, state-driven widgets and live REST data
- Backend, database, payment server logic and DevOps remained outside the internship scope

Problem Statement

Why Beast OS was conceived

Fragmented preparation

Students rely on printed material, generic apps and disconnected practice tests.

No unified progress view

Daily activity, streaks and syllabus progress are not visible in one place.

Weak-topic discovery

Students lack an at-a-glance breakdown of board readiness by subject and chapter.

Unclear monetisation

Subscription screens can be cluttered or unclear about plan inclusions.

Limited internal visibility

Organisations need a real-time view of subscriber activity, credit usage and plan distribution.

**Beast OS: one cohesive front-end system for students
+ admins**

Objectives & Scope

What was designed, built and integrated during the internship

Project Objectives

- Unified Student Dashboard with daily challenge, streak and quick access
- AI Tutor chat interface with credit-based usage
- Configurable Quiz Generator for subject/chapter/quiz length
- Command Centre for board readiness, subject health and biggest gaps
- Subscription Store with Monthly/Yearly Pro plans
- Practical experience in component architecture, responsive design, UI state and data visualisation

Included in Scope

React component design • Tailwind responsive layout • client-side state handling • dynamic widgets • REST API integration • production-ready web UI

Outside Scope

Backend/API development • database schema design • server-side payment-gateway logic • infrastructure / DevOps ownership

Technology Stack

Tools and technologies used within the Web Developer scope

React.js + Tailwind CSS

Component-based, responsive front-end UI

Chart.js / Recharts

Daily Active Credits and Subscription Plan Distribution charts

React Hooks + REST APIs

State handling and live data fetching

Lucide + Custom Cards

Consistent iconography and reusable visual components

Supabase / PostgreSQL

Consumed backend source of live dashboard data

Razorpay API

Consumed for Subscription Store upgrade actions

System Architecture

Decoupled client-server architecture with a reusable presentation layer

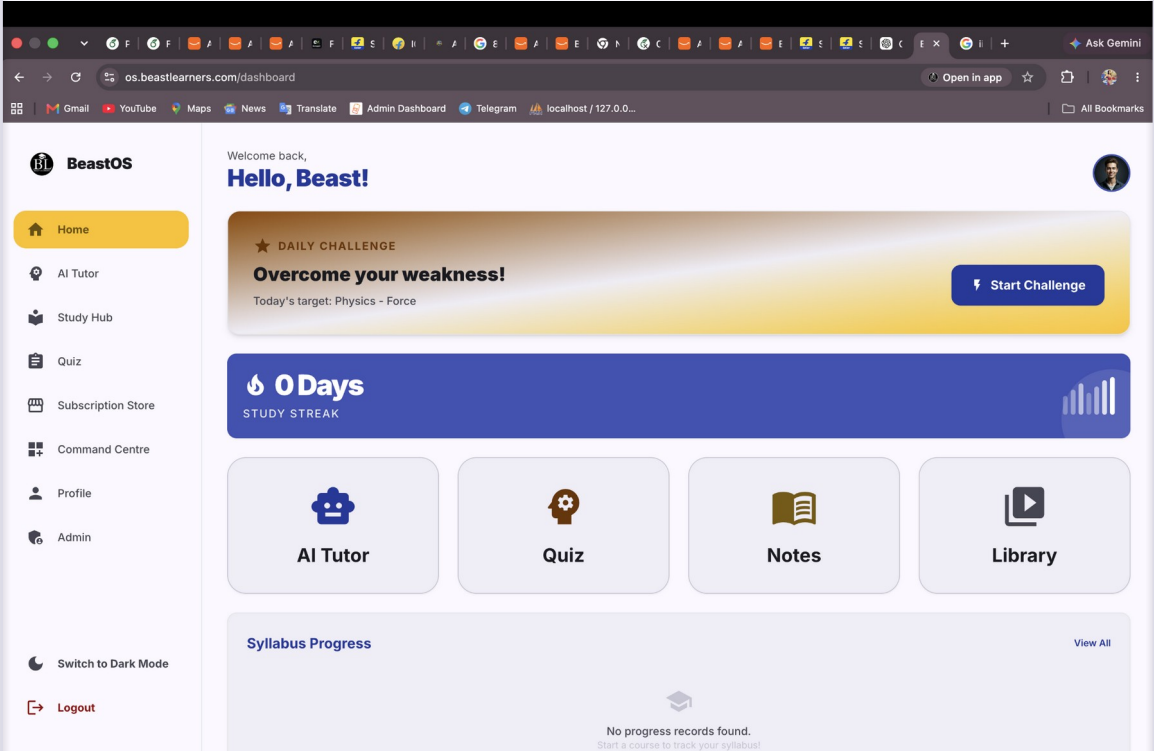


Design Principle

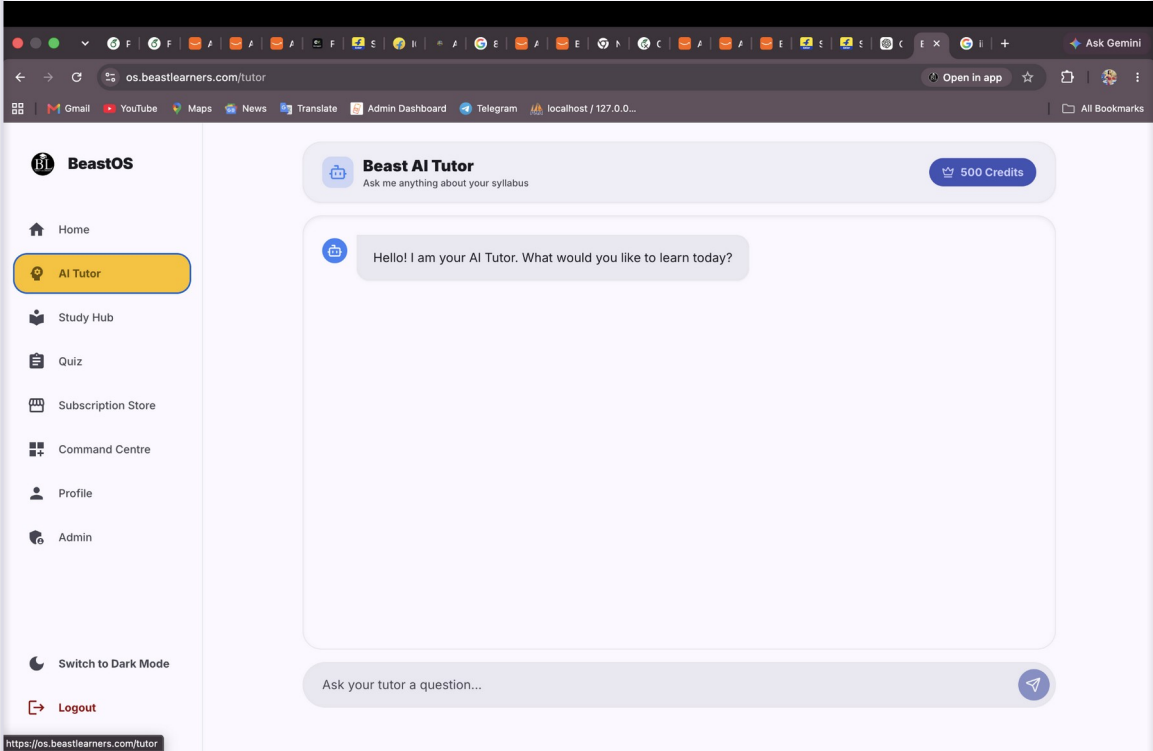
Business-critical values such as readiness %, credit balances and subscription status are computed server-side; the client renders the trusted source of truth.

Student Experience

Home Dashboard & AI Tutor



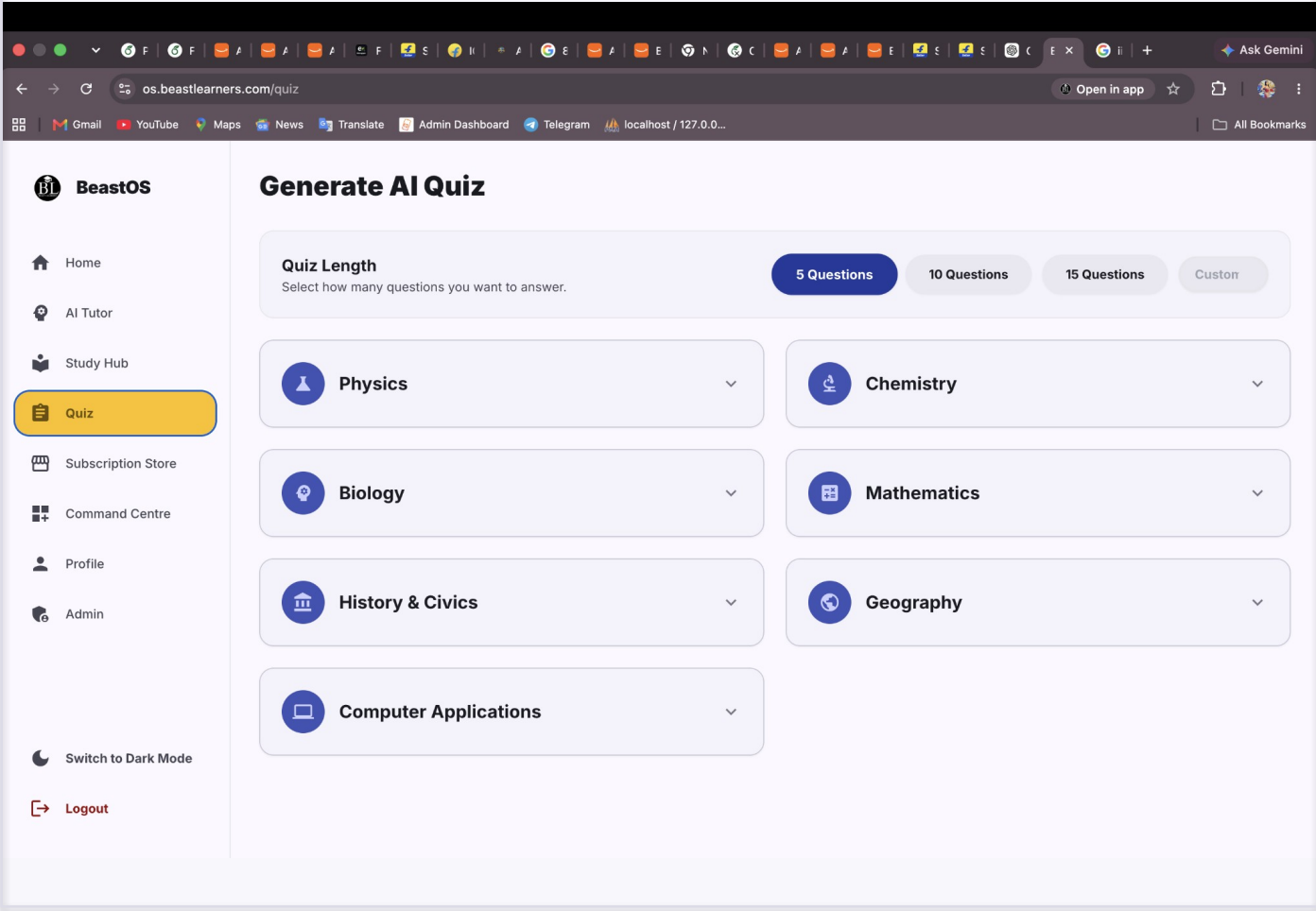
HOME DASHBOARD



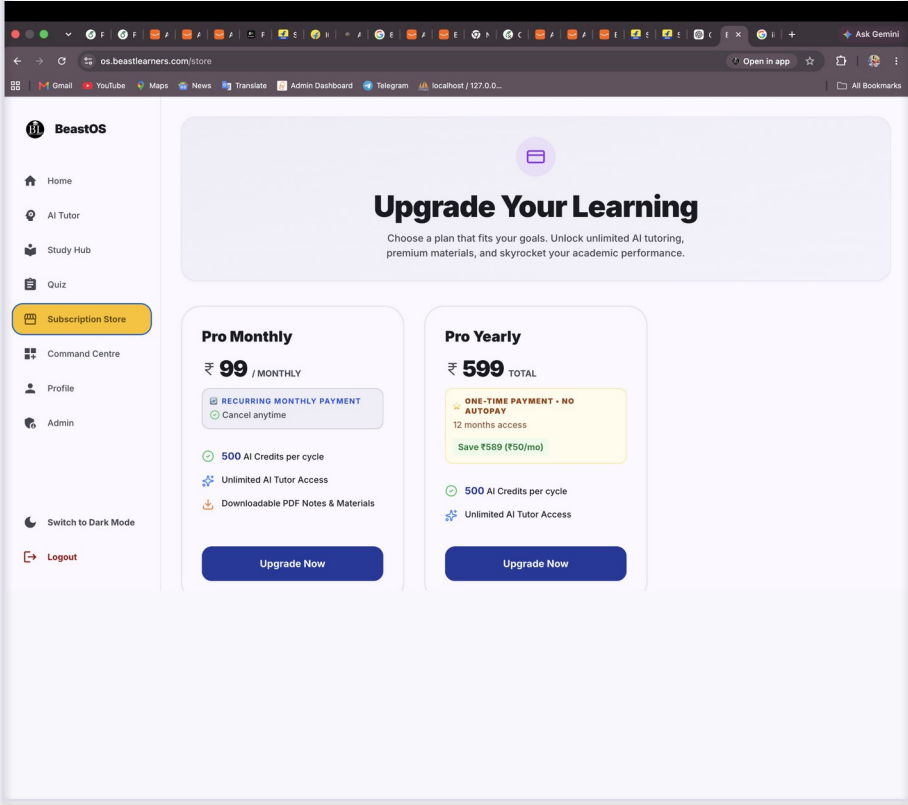
BEAST AI TUTOR

Learning & Monetisation

Quiz Generator & Subscription Store



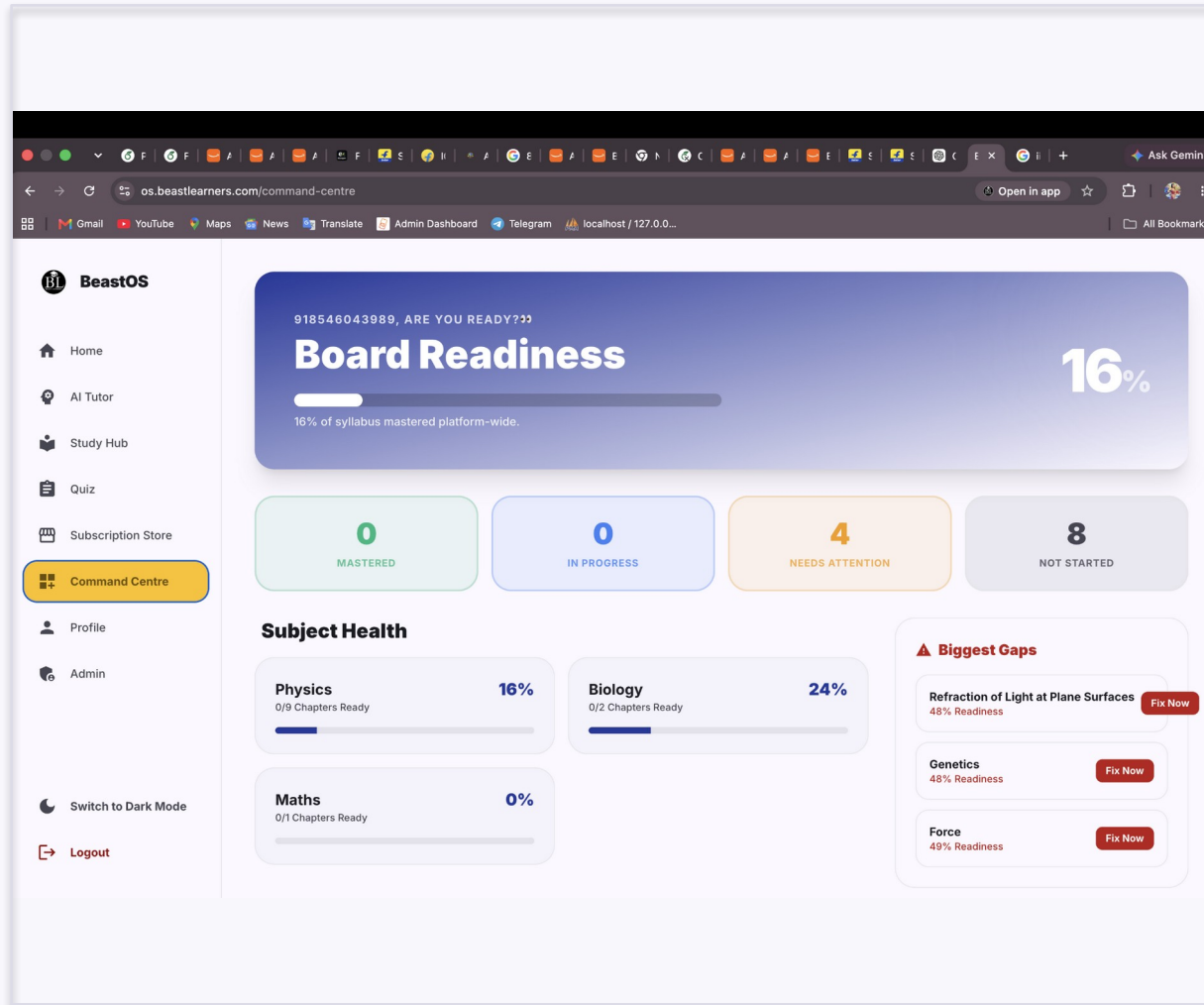
AI QUIZ GENERATOR



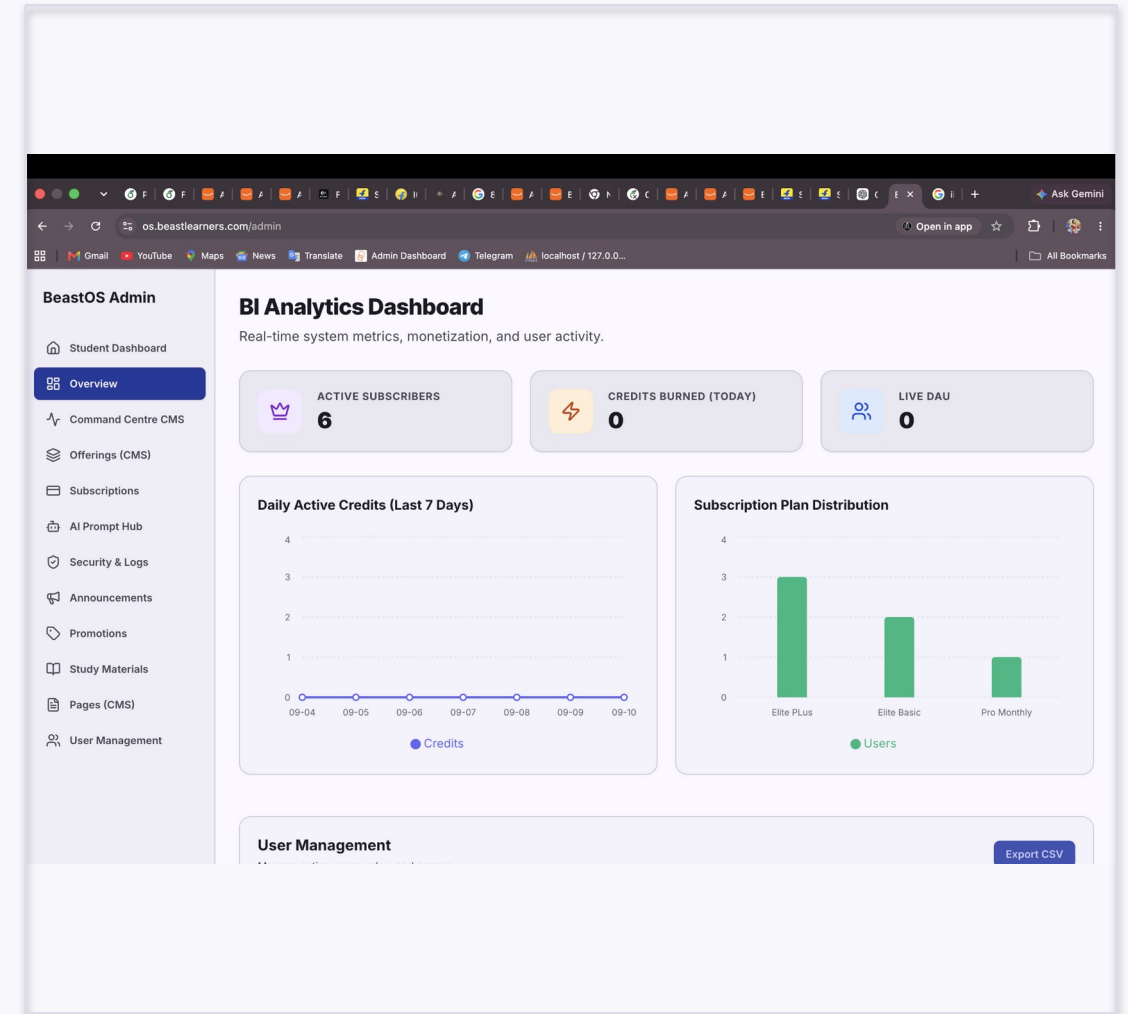
SUBSCRIPTION STORE

Analytics & Decision Support

Command Centre + Admin BI Dashboard



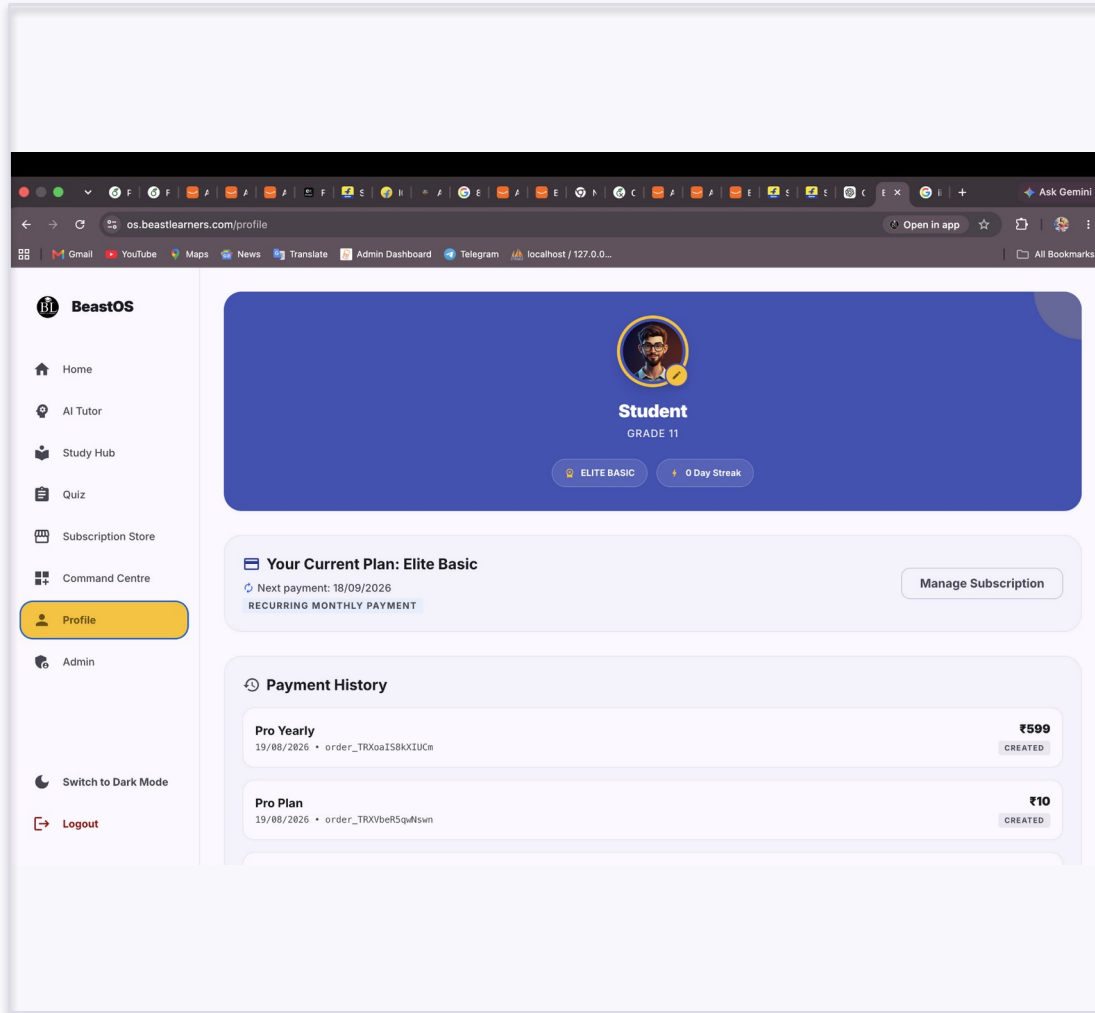
COMMAND CENTRE — BOARD READINESS



ADMIN BI ANALYTICS

Supporting UI & Development Methodology

Profile screen and agile, screen-by-screen implementation



1. Wireframe

Define layout, hierarchy and interaction flow.

2. Componentise

Build reusable cards, navigation, progress and chart components.

3. Connect

Consume REST endpoints and render live platform data.

4. Ship Iteratively

Validate responsive behaviour, empty states and UI consistency.

Technical Challenges & Solutions

Key engineering decisions during implementation

Reusable components

Shared state

Empty / zero-data states

Explicit UI

Balanced quiz grid

Responsive

Live dashboard values

React hooks

Growing gap lists

Scroll/overflow

Outcomes & Learning

Impact of the internship and skills developed

Production-Facing Outcomes

- Five major front-end modules shipped as part of Beast OS
- Home dashboard created a single motivating entry point for study
- AI Tutor and Quiz Generator made learning tools easier to discover and use
- Command Centre visualised board readiness and weakest topics
- Subscription Store and Admin BI Dashboard supported monetisation and internal monitoring

Professional Skills Gained

- Component-based front-end architecture
- Responsive UI engineering with Tailwind CSS
- Data visualisation using chart libraries
- Consuming live REST APIs and rendering changing data
- Handling real product constraints, empty states and reusable UI patterns
- Stronger readiness for professional front-end engineering

Future Scope & Conclusion

Next opportunities identified in the project report

Command Centre

Add trend charts showing readiness improvement over time.

Quiz Generator

Support drag-and-drop chapter reordering for more flexible assessment setup.

Admin BI

Add exportable analytics with date-range filtering.

Beast OS transformed the internship into hands-on experience building a real, data-driven EdTech front end.

THANK YOU

Aditya Raj • Web Developer Intern • Beast Learners

BEAST OS